Role of diet and exercise in the management of hyperinsulinemia and associated atherosclerotic risk factors.

Hyperinsulinemia, hypertension, hypertriglyceridemia and obesity are independent risk factors for coronary artery disease and are often found in the same person. This study investigated the effects of an intensive, 3-week, dietary and exercise program on these risk factors. The group was divided into diabetic patients (non-insulin-dependent diabetes mellitus [NIDDM], n = 13), insulin-resistant persons (n = 29) and those with normal insulin, less than or equal to 10 microU/ml (n = 30). The normal groups had very small but statistically significant decreases in all of the risk factors. The patients with NIDDM had the greatest decreases. Insulin was reduced from 40 +/- 15 to 27 +/- 11 microU/ml, blood pressure from 142 +/- 9/83 +/- 3 to 132 +/- 6/71 +/- 3 mm Hg, triglycerides from 353 +/- 76 to 196 +/- 31 mg/dl and body mass index from 31.1 +/- 4.0 to 29.7 +/- 3.7 kg/m2. Although there was a significant weight loss for the group with NIDDM, resulting in the decrease in body mass index, 8 of 9 patients who were initially overweight were still overweight at the end of the program, and 5 of the 8 were still obese (body mass index greater than 30 kg/m2), indicating that normalization of body weight is not a requisite for a reduction or normalization of other risk factors. Insulin was reduced from 18.2 +/- 1.8 to 11.6 +/- 1.2 microU/ml in the insulin-resistant group, with 17 of the 29 subjects achieving normal fasting insulin (less than 10 microU/ml).(ABSTRACT TRUNCATED AT 250 WORDS)